SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA06 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 40%

QUESTIONS: 15 Total Marks: 25

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I Savitha R / 192421368 (Name / Reg No) certify that this submission is my original work and
that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SAVITHA R

Annexure A

Short Answer Test

1. A factory has three boxes containing coloured balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

Concept Used: This problem is solved using Bayes' Theorem, which is used to determine the probability of an event based on prior information.

Given Box A: 6 Red, 4 Blue, 2 Green,Total balls = 12

$$\boldsymbol{P}(\textbf{Blue} \mid \boldsymbol{A}) = \frac{\mathbf{4}}{\mathbf{12}} = \frac{\mathbf{1}}{\mathbf{3}}$$

Box B: 5 Red, 7 Blue, 3 Green, Total balls = 15

$$P(\text{Blue} \mid B) = \frac{7}{15}$$

Box C: 8 Red, 2 Blue, 5 Green, Total balls = 15

$$P(\text{Blue} \mid C) = \frac{2}{15}$$

Since Box B is twice as likely to be selected, $P(A) = \frac{1}{4},\ P(B) = \frac{1}{2},\ P(C) = \frac{1}{4}$

Bayes' Theorem

$$P(C \mid Blue) = \frac{P(Blue \mid C) \times P(C)}{P(Blue)}$$

First calculate $P(Blue) = P(Blue \mid A)P(A) + P(Blue \mid B)P(B) + P(Blue \mid C)P(C)$

$$= \frac{1}{3}\left(\frac{1}{4}\right) + \frac{7}{15}\left(\frac{1}{2}\right) + \frac{2}{15}\left(\frac{1}{4}\right)$$

$$= \frac{1}{12} + \frac{7}{30} + \frac{1}{30} = \frac{7}{20}$$

Now, $P(C \mid Blue) = \dfrac{\frac{2}{15}\times\frac{1}{4}}{\frac{7}{20}} = \dfrac{2}{21}$

Final Answer $\boxed{P(\text{Box C} \mid \text{Blue}) = \dfrac{2}{21} = 0.0952}$

Probability = 9.52%

Conclusion: After observing a blue ball, the probability that it came from Box C is 9.52%.

2. **A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?**

Concept Used

This problem uses Bayes' Theorem.

Given: Section A: Total = 10

$$P(Blue \mid A) = \frac{3}{10}$$

Section B: Total = 15

$$P(Blue \mid B) = \frac{6}{15} = \frac{2}{5}$$

Section C: Total = 15

$$P(Blue \mid C) = \frac{2}{15}$$

Selection probabilities:

$$P(A) = \frac{1}{4},\ P(B) = \frac{1}{2},\ P(C) = \frac{1}{4}$$

Using Bayes' Theorem

$$P(C \mid Blue) = \frac{P(Blue \mid C)P(C)}{P(Blue)}$$

Calculate

$$P(Blue) = \frac{3}{10} \times \frac{1}{4} + \frac{2}{5} \times \frac{1}{2} + \frac{2}{15} \times \frac{1}{4}$$
$$= \frac{3}{40} + \frac{1}{5} + \frac{1}{30} = \frac{37}{120}$$

Therefore,

$$P(C \mid Blue) = \frac{\frac{2}{15} \times \frac{1}{4}}{\frac{37}{120}} = \frac{4}{37}$$

Final Answer

$$\boxed{P(\text{Section C} \mid \text{Blue}) = \frac{4}{37} = 0.1081}$$

Probability = 10.81%

Conclusion: If the selected car is blue, the probability that it came from

Section C is 10.81%.

3.In a real-world medical image classification task for detecting breast cancer, a deep- learning model integrated with the K-Nearest Neighbours algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness inmedical diagnosis.

Concept Used: A Confusion Matrix is used to evaluate the performance of a binary classifier.

Given

- **Total samples per class = 1800**
- **Test data = 20%**
- **TP = 210**
- **TN = 190**
- **FP = 30**

Step 1: Find FN

Positive test samples

$$\mathbf{1800 \times 20\% = 360}$$

Since

$$\boldsymbol{TP + FN = 360}$$
$$\boldsymbol{FN = 360 - 210 = 150}$$

Step 2: Calculate Performance Metrics

Accuracy

$$\boldsymbol{Accuracy = \frac{TP + TN}{TP + TN + FP + FN}}$$

$$= \frac{210 + 190}{210 + 190 + 30 + 150} = \frac{400}{580} = 68.97\%$$

Precision

$$Precision = \frac{TP}{TP + FP}$$
$$= \frac{210}{240} = 87.50\%$$

Sensitivity (Recall)

$$Sensitivity = \frac{TP}{TP + FN}$$
$$= \frac{210}{360} = 58.33\%$$

Specificity

$$Specificity = \frac{TN}{TN + FP}$$
$$= \frac{190}{220} = 86.36\%$$

Final Answer

Metric	Value
False Negative (FN)	150
Accuracy	68.97%
Precision	87.50%
Sensitivity	58.33%
Specificity	86.36%

Conclusion

The classifier has high precision (87.50%) and high specificity (86.36%), indicating reliable prediction of positive cases and healthy patients. However, the low sensitivity (58.33%) means many cancer cases are missed. For medical diagnosis, improving sensitivity is essential.

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs:

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4 (b)

Epoch 2: Training Loss = 2.5, Validation Loss = 2.2 **(c)**

Epoch 3: Training Loss = 2.2, Validation Loss = 2.0 **(d)**

Epoch 4: Training Loss = 2.0, Validation Loss = 2.1 **(e)**

Epoch 5: Training Loss = 1.8, Validation Loss = 2.3 **(f)**

Epoch 6: Training Loss = 1.6, Validation Loss = 2.6

(g)Epoch 7: Training Loss = 1.5, Validation Loss = 2.9 **(h)**

Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i)Identify the epoch at which the model begins to overfit.

ii) Suggest two effective techniques to reduce overfitting in this model.

iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Concept Used: Overfitting occurs when the training loss continues to decrease but the validation loss starts increasing.

(i) Identify the Overfitting Epoch

Epoch	Training Loss	Validation Loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6

Epoch	Training Loss	Validation Loss
7	1.5	2.9
8	1.3	3.2

The validation loss reaches its minimum at Epoch 3 and starts increasing from Epoch 4, while the training loss keeps decreasing.

Answer: Overfitting begins at Epoch 4.

(ii) Two Techniques to Reduce Overfitting

1. Early Stopping – Stop training when validation loss starts increasing.
2. Dropout Regularization – Randomly deactivate neurons during training to prevent memorization.

(iii) Effect of Overfitting

Overfitting causes the model to memorize the training data rather than learning general patterns. As a result, the model performs well on training data but poorly on unseen test data, reducing its generalization ability.

Conclusion: The model begins overfitting at Epoch 4. Applying Early Stopping and Dropout can improve performance on unseen data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs:

(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%

(c) Epoch 3: Training Accuracy = 70%,

Validation Accuracy = 68%

(d) **Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%**

(e) **Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%**

(f) **Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%**

(g) **Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%**

(h)**Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%**

i)**Identify the epoch at which the model starts overfitting.**

ii) **Explain why validation accuracy decreases despite training accuracy increasing.**

iii) **Suggest two methods to improve validation performance.**

Concept Used: Overfitting occurs when training accuracy keeps increasing while validation accuracy starts decreasing.

(i) Identify the Overfitting Epoch

Epoch	**Training Accuracy**	**Validation Accuracy**
1	**60%**	**58%**
2	**65%**	**63%**
3	**70%**	**68%**

Epoch	Training Accuracy	Validation Accuracy
4	75%	72%
5	80%	73%
6	85%	72%
7	88%	70%
8	92%	68%

Validation accuracy reaches its highest value (73%) at Epoch 5 and decreases afterward.

Answer: Overfitting starts at Epoch 6.

(ii) Why Does Validation Accuracy Decrease?

Training accuracy continues to improve because the model memorizes the training data. However, it fails to generalize to unseen validation data, causing validation accuracy to decrease.

(iii) Two Methods to Improve Validation Performance

1. Early Stopping
2. Data Augmentation (or Dropout/L2 Regularization)

Conclusion: The model starts overfitting at Epoch 6. Using Early Stopping and Data Augmentation can improve validation accuracy and increase the model's generalization performance.